Native inverse feeds pivot-row normalization

Task T-1489. 3 November 2026.

Native inverse	Returned narrow word, canonical and inverse
Composition	Pivot one + prefix zero + equation equivalence
Kernel checks	3 theorems + 8 controls + 11 axiom readbacks
Retained premises	Prime bound, canonical row, nonzero, prefix

Exact inverse-value premise discharged; native array-loop bridge remains. T-1489.

Three composition theorems passed with eight controls and eleven named axiom readbacks. Normalization now takes the concrete returned narrow native inverse value, rather than an arbitrary unit supplied with inverse and canonicality assumptions. The source-level construction is ready for independent review.

Construction. returnedUnit applies RecoverNarrowPow.inverseGenerator to the pivot word. returned_unit_contract derives its canonical range from generator_product and its decoded field inverse from generator_decode. normalize_contract derives decoded pivot nonzero from native nonzero plus canonical input, then instantiates the reviewed word_inverse_contract. It proves canonical output, unchanged zero prefix, pivot exactly one, and equivalence of augmented row equations for every four-coordinate field vector. normalized_residual also states the exact residual product by the pivot inverse. No desired output equation appears among the premises.

Retained premises. The UInt64 modulus is prime and at most2^32; every augmented-row entry is canonical; the pivot word is nonzero; entries before the current column are zero. The theorem derives1<p from primality. The inverse value is the literal narrow multiplication/low-bit/unconditional-square/right-shift model with64-step fuel, as reviewed in the predecessor. Counter increments and overflow are excluded. Pivot search, swapping, establishment of prefix zeros, concrete mutable-array loops and the full solve4 return remain separate source bridges.

Source fidelity. Native recover_boundary_impl.rs lines102-105 capture inv(mat[col][col],p), then multiply each augmented suffix entry by that unit. The composition models that fixed value and the existing finite row function. Native source SHA56a6bda57c9e8402e6ec80f3d6fd5ba5ba3308aae23c1c42480b7023b01a7183 is bound by both predecessor packets. Narrow and Normalization sources were copied byte-for-byte and recompiled during the final check. This is manual source correspondence, not verified extraction of Rust or machine-code semantics.

Controls. At p7 the row[0,2,4,0,6] returns unit4 and normalizes to[0,1,2,0,3]. A separate premise witness and theorem application establish nonvacuity. Additional fixtures cover characteristic2, the last pivot and augmented RHS, zero-pivot failure, and a missing-prefix counterexample where the equation is not preserved. The deliberately false normalized pivot equals2 was rejected by decide. All eleven named declarations use only propext, Classical.choice and Quot.sound; no sorryAx is accepted.

Attempts and failures. The initial two95-second attempts were authorized prospectively. Attempt1 compiled Narrow then failed on a missing RowFold import in the isolated normalization closure,30.622275 seconds. Exact reviewed dependency files were added. Attempt2 compiled all three composition theorems but failed control parsing because arrow immediately followed by row was tokenized as arrow-r notation; the resulting sorry warning was rejected and axiom checking was not reached,22.418847 seconds. A separately authorized third95-second attempt changed only control whitespace and passed the complete check in36.351181 seconds. All failures and source versions are retained. Total89.392303 seconds against285 seconds authorized ceilings. No further invocation ran.

Verification boundary. The final six Lean calls used one worker,8GiB address space,3500MiB Lean heap,16MiB internal stack and10MiB hard OS stack. All child processes were reaped;28,808 dependency, toolchain and input hashes were unchanged. Shared dependency binaries remain trusted; no separate binary/kernel rechecker ran. An exact source archive retains local dependency snapshots and recursive inventories. No scientific trajectories, cloud operations or timing comparison were executed.

Next review. A non-author should inspect whether the generator and normalization pivot indices coincide, check that derived unit premises are actually discharged, replay characteristic2 and missing-prefix controls, and verify the exact source-to-row-function boundary. The author previously produced normalization and independently reviewed the suffix component; this packet is a producer composition, not its own independent review.